

Question number	Answer	Mark
3(a)	The only correct answer is D <i>A is incorrect because TMV is an RNA virus</i> <i>B is incorrect because TMV is an RNA virus</i> <i>C is incorrect because TMV does not have an envelope</i>	(1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	8.7% of the 8.3×10^{12} calculated (1) 7.6×10^{12} (1) OR - percentage recycled calculated (1) 7.6×10^{12} (1)	$0.7221 \times 10^{12} / 7.221 \times 10^{11}$ 91.3 Bald answer of $7.6 \times 10^{12} = 2$ marks Bald answer of $7.5779 \times 10^{12} / 76 \times 10^{11} / 0.7221 \times 10^{12} / 7.221 \times 10^{11} / 91.3 = 1$ mark	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	An explanation that includes the following points: - because they {produce / release} enzymes (1) - that can {break / hydrolyse} bonds within the plastic (1)		(2)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that includes the following points: • because breakdown products {taken up / absorbed} (by the bacteria) (1) • respiration produces carbon dioxide (1) • (bacterial) {biomass / stored material} is {(other) organic molecules / converted breakdown products / glycogen / polysaccharide / protein / lipid} (1)	ACCEPT converting what is taken up into {another molecule / glucose / named (simple) molecule} DO NOT ACCEPT molecules not found in bacteria e.g. starch, chlorophyll	(3)